

BARUUN-URT, Mongolia

WMO: 443050

Lat: 46.6712N

Lon: 113.2845E

Elev: 981

StdP: 90.08

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.7	-28.8	-34.2	0.2	-30.1	-32.1	0.2	-28.0	11.6	-15.9	9.7	-15.4	2.7	320	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.7	18.4	30.4	17.6	28.5	16.8	20.5	29.0	19.4	27.2	18.4	25.3	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	14.6	23.2	17.0	13.7	22.1	16.0	12.8	21.7	64.2	28.8	60.0	27.2	56.4	25.4	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.5	10.4	9.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-33.0	35.9	2.1	2.5	-34.5	37.7	-35.7	39.1	-36.9	40.5	-38.5	42.3
			-33.0	21.9	2.3	2.1	-34.7	23.4	-36.0	24.6	-37.3	25.9	-39.0	27.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-21.0	-15.9	-5.8	4.7	12.2	18.6	21.5	19.5	12.9	3.1	-9.2	-18.0		
	DBStd	15.63	4.74	5.63	6.37	5.22	4.93	4.10	3.27	3.61	4.74	5.12	6.52	4.67		
	HDD10.0	4067	959	726	488	173	33	1	0	0	24	219	576	868		
	HDD18.3	6206	1218	959	747	410	197	45	6	28	172	472	826	1126		
	CDD10.0	1144	0	0	0	13	102	259	357	296	111	6	0	0		
	CDD18.3	240	0	0	0	0	8	54	104	66	8	0	0	0		
	CDH23.3	2470	0	0	0	6	152	608	966	629	109	1	0	0		
	CDH26.7	875	0	0	0	1	43	222	372	213	24	0	0	0		
	WSAvg	3.8	3.2	3.3	4.1	5.0	5.1	4.1	3.6	3.4	3.7	3.7	3.3	3.2		
	PrecAvg	193	2	2	3	5	15	37	58	44	21	5	3	3		
(15) Precipitation	PrecMax	373	14	6	16	24	72	103	165	107	86	20	11	49		
	PrecMin	104	0	0	0	0	1	2	5	6	0	0	0	0		
	PrecStd	60	2	2	3	6	14	25	36	25	19	5	3	7		
	PrecStd	60	2	2	3	6	14	25	36	25	19	5	3	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.6	1.8	14.5	23.6	31.1	34.2	36.1	34.6	29.5	21.0	8.8	-3.5		
		MCWB	-8.2	-1.5	4.7	10.1	13.9	17.8	20.6	19.6	16.2	10.7	2.6	-5.4		
	2%	DB	-9.7	-2.1	10.2	19.8	27.2	31.6	33.0	31.2	26.5	16.8	5.3	-7.2		
		MCWB	-10.9	-4.3	2.9	8.9	13.4	17.3	19.4	18.0	14.7	8.4	0.8	-8.5		
	5%	DB	-12.0	-5.2	6.8	16.8	24.5	29.2	30.6	28.9	23.7	13.7	2.4	-9.4		
		MCWB	-12.8	-6.7	1.1	7.0	12.0	16.5	18.7	16.9	13.5	6.6	-1.2	-10.4		
	10%	DB	-14.1	-7.5	3.7	14.1	21.9	26.8	28.4	26.8	21.2	11.3	-0.3	-11.3		
		MCWB	-14.8	-8.7	-0.8	5.8	10.9	15.5	17.9	16.3	12.4	5.0	-3.0	-12.1		
	0.4%	WB	-8.1	-1.3	5.6	11.7	16.8	20.7	22.9	21.4	18.6	11.6	3.2	-5.3		
		MCDB	-6.7	1.3	13.6	20.8	27.0	30.9	32.7	30.5	25.2	19.7	7.9	-3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-10.8	-4.4	3.0	9.4	14.4	18.9	21.1	19.7	15.9	8.6	0.8	-8.4		
		MCDB	-9.8	-2.2	9.6	18.4	24.3	28.2	29.2	28.4	23.3	15.3	5.1	-7.3		
	5%	WB	-12.8	-6.7	1.2	7.6	12.9	17.6	20.0	18.6	14.5	7.1	-1.1	-10.4		
		MCDB	-12.0	-5.2	6.5	16.1	22.3	25.9	27.5	25.8	21.9	13.2	2.2	-9.5		
	10%	WB	-14.8	-8.7	-0.6	5.9	11.5	16.4	19.0	17.6	13.1	5.3	-2.9	-12.0		
		MCDB	-14.2	-7.5	3.5	13.8	20.6	24.2	25.9	23.9	19.8	10.8	-0.5	-11.3		
	5% DB	MDBR	8.3	10.3	11.4	12.6	13.3	11.6	10.9	11.6	12.3	11.4	9.8	8.2		
		MCDBR	10.0	11.9	15.0	17.0	17.3	15.1	14.1	14.6	15.4	14.7	12.2	9.7		
		MCWBR	9.1	10.0	9.6	9.9	8.7	6.4	5.7	6.1	7.7	8.9	8.9	8.7		
		MCWBR	10.0	11.8	14.3	16.0	15.2	13.2	12.1	12.3	13.6	13.8	12.0	9.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.1	10.1	9.4	10.1	8.8	6.7	6.0	6.2	7.5	9.0	8.9	8.7		
		MCWBR	9.1	10.1	9.4	10.1	8.8	6.7	6.0	6.2	7.5	9.0	8.9	8.7		
	Clear-Sky Solar Irradiance	taub	0.253	0.275	0.315	0.373	0.436	0.453	0.411	0.379	0.340	0.305	0.270	0.256		
		taud	2.328	2.287	2.249	2.135	1.992	2.056	2.251	2.310	2.354	2.380	2.359	2.302		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	853	900	907	878	830	817	849	864	871	852	821	799		
		Edh at Noon	73	94	115	143	171	162	132	119	103	86	70	67		
(44) All-Sky Solar Radiation	RadStd	RadAvg	2.06	3.20	4.65	5.87	6.63	6.59	6.29	5.81	4.75	3.37	2.16	1.62		
		RadStd	0.13	0.17	0.15	0.22	0.35	0.44	0.28	0.31	0.27	0.19	0.13	0.12		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-2.04	N/A	-1.26	-1.26	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	-1.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page